OPERATION DIM STAR: EXOPLANET DETECTION & HABITABILITY ANALYSIS FOR REBEL ALLIANCE BASE RELOCATION

STAC WARS Problem Statement 3. Technical Report

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ABSTRACT

After the Death Star was destroyed Imperial forces are still actively looking for Rebel groups over known space. To find a place to operate Alliance Intelligence did a detailed study of the brightness of stars for three possible systems: STAC-1, STAC-2 and STAC-3. Using the way changes when a planet passes in front of a star the way planets move around stars and models of where life could exist we found possible exoplanets checked their size and other features and saw if they could support people. Our results show that STAC-3b is the place for the Rebels to move to right away.

I. EXOPLANET DETECTION & LIGHT-CURVE ANALYSIS

When a planet moves directly between its star and an observer it blocks a small part of the stars light making the brightness go down a little. The amount of the drop is called the transit depth. We looked at the brightness data for 200 days and found these things for each system:

A. STAC-1 System Analysis

• Transit Detection: We saw drops in brightness on day 3 8 13 and 18.

• Orbital Period Calculation: The time between each drop is always 5.0 days. So the orbital period is 5.0 days.

• Transit Depth Calculation: The normal brightness is 1.0000 and the brightness during the drop is 0.9970. The drop is 0.30 percent.

B. STAC-2 System Analysis

• Transit Detection: We saw drops in brightness on day 20 100 and 180.

• Orbital Period Calculation: The time between each drop is 80.0 days. So the orbital period is 80.0 days.

• Transit Depth Calculation: The normal brightness is 1.0000 and the brightness during the drop is 0.9990. The drop is 0.10 percent.

C. STAC-3 System Analysis

• Transit Detection: We saw drops in brightness on day 20 60 100 140 and 180.

• Orbital Period Calculation: The time between each drop is always 40.0 days. So the orbital period is 40.0 days.

• Transit Depth Calculation: The normal brightness is 1.0000 and the brightness during the drop is 0.9990. The drop is 0.10 percent.

II. PLANETARY RADIUS & PHYSICAL CLASSIFICATION

If a planet is round and passes in front of a round star the drop in brightness shows the size of the planet. We used the size of the sun and Earth to find the size of each planet:

A. STAC-1b Radius & Classification

1. The size of the star is 1.00 times the size of the sun.

2. The size of the planet is 38,121 kilometers.

3. The size of the planet is 5.98 times the size of Earth.

4. The planet is the size of Neptune.

B. STAC-2b Radius & Classification

1. The size of the star is 0.80 times the size of the sun.

2. The size of the planet is 17,607 kilometers.

3. The size of the planet is 2.76 times the size of Earth.

4. The planet is like a Neptune or a big Earth.

C. STAC-3b. Classification

1. The size of the star is 0.50 times the size of the sun.

2. The size of the planet is 11,005 kilometers.

3. The size of the planet is 1.73 times the size of Earth.

4. The planet is a Super-Earth.

III. PLANETARY. Keplerian DYNAMICS

We used a rule called Keplers Third Law to find out how far the planet is from the star. The rule uses the time it takes to orbit and the size of the star.

A. STAC-1b Orbit

1. The time it takes to orbit is 0.01369 years. The star is the size as the sun.

2. The distance is 0.0572 AU.

3. The distance is 8,561,400 kilometers.

4. The planet is very close to the star.

B. STAC-2b Orbit

1. The time it takes to orbit is 0.21903 years. The star is 0.80 times the size of the sun.

2. The distance is 0.3373 AU.

3. The distance is 50,462,800 kilometers.

4. The planet is close to the star like Mercury.

C. STAC-3b Orbit

1. The time it takes to orbit is 0.10951 years. The star is 0.50 times the size of the sun.

2. The distance is 0.1817 AU.

3. The distance is 27,188,300 kilometers.

4. The planet is not too close to the star.

IV. HABITABILITY EVALUATION & CANDIDATE SELECTION

We looked at how close the planet's to the area where life could exist.

A. Habitable Zone Boundaries vs Orbital Distances

System	**Luminosity (L_⊙)**	**d_HZ Center (AU)**	**Actual Orbit a (AU)**	**HZ Match Evaluation**
STAC-1b	1.00	1.000 AU	0.057 AU	Way inside HZ (Extremely Hot)
STAC-2b	0.40	0.632 AU	0.337 AU	Inside HZ inner edge (Overheated)
STAC-3b	0.08	0.283 AU	0.182 AU	Optimal Habitable Zone Candidate

B. Candidate Selection & Supporting Evidence

The best choice is STAC-3b. Here are three reasons:

1. The planet is in the place for life. The star is an one and the planet is in the area where water could exist.

2. The planet is a place, not a gas planet. This means it can support bases.

3. The planet is hard to see from away. This makes it a good place to hide.

C. Recommended Atmospheric Follow-Up

To check if the planet has air and if life can exist we should look at the planet during a transit. We can see the air by looking at the light that goes through the planets atmosphere.

V. COMPREHENSIVE SYSTEM COMPARISON TABLE

Target System	**Period P**	**Depth δ**	**Radius R_p**	**Classification**	**Orbit a**	**Habitability**
STAC-1b	5.0 days	0.30%	5.98 R_⊕	Neptune-sized	0.057 AU	Uninhabitable
STAC-2b	80.0 days	0.10%	2.76 R_⊕	Mini-Neptune	0.337 AU	Too Hot
STAC-3b	40.0 days	0.10%	1.73 R_⊕	Super-Earth	0.182 AU	**HIGHLY SUITABLE**